volume, you pay for 100 GiB, even if you have no data on the volume. If you use EBS Magnetic HDD (standard) volumes, you must also pay for every I/O operation you perform. Provisioned IOPS SSD (io1) volumes are additionally charged based on the provisioned IOPS. Use the AWS Simple Monthly Calculator at https://calculator.aws/ to determine how much your storage setup will cost.

We advise you to use general-purpose (SSD) volumes as the default. If your workload requires more IOPS, then go with provisioned IOPS (SSD). You can attach multiple EBS volumes to a single EC2 instance to increase overall capacity or for additional performance.

8.1.4 Backing up your data with EBS snapshots

EBS volumes replicate data on multiple disks automatically and are designed for an annual failure rate (AFR) of 0.1% and 0.2%. This means on average you should expect to lose 0.5–1 of 500 volumes per year. To plan for an unlikely (but possible) failure of an EBS volume, or more likely a human failure, you should create backups of your volumes regularly. Fortunately, EBS offers an optimized, easy-to-use way to back up EBS volumes with EBS snapshots. A *snapshot* is a block-level incremental backup. If your volume is 5 GiB in size, and you use 1 GiB of data, your first snapshot will be around 1 GiB in size. After the first snapshot is created, only the changes will be persisted, to reduce the size of the backup. EBS snapshots are charged based on how many gigabytes you use.

You'll now create a snapshot using the CLI. Before you can do so, you need to know the EBS volume ID. You can find it as the